

Part One: Foundations, Regression, and Generalized Linear Models

An Introduction to the Chapters and How They Fit Together

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1 What Part One Offers

Quantitative Methods for Social and Behavioral Research develops the statistical understanding needed to read empirical studies, interpret their results, and conduct quantitative research. Part One is where the book begins: fourteen numbered chapters that start with what a study asks and how it produces its evidence, and end with regression models for a range of outcomes. It is written for postgraduate students and applied researchers. A mean, an interval, or a regression coefficient becomes useful when a reader understands what it describes, how it was obtained, and which question it helps answer.

Some introductory familiarity with statistics and algebra is useful, and little else is assumed. Each chapter explains the notation it uses and develops its calculations in worked steps, so the algebra arrives with the idea that needs it. Calculus and proof-based mathematical statistics are not prerequisites. Chapters 1–4 can be read without running R or SPSS: Chapter 1 introduces the research journey, and Chapters 2–4 teach statistics directly, through explanations, figures, and worked calculations. Computation joins the statistics in Chapters 5 and 6.

Results are rounded to four decimals from full-precision computation, with trailing zeros dropped (3.1 rather than 3.1000), unless a source's own precision applies.

2 How the Chapters Build on Each Other

Nine chapters are available, and they form one route. Chapter 1, *The Quantitative Research Journey*, orients the reader in questions, evidence, design, and interpretation. Chapter 2, *Data, Distributions, and Variation*, develops records, distributions, and variation across repeated samples. Chapter 3, *Estimates, Uncertainty, and Statistical Tests*, develops estimation, intervals, and tests. Chapter 4, *Group Comparisons with Analysis of Variance*, develops comparisons among several groups. Chapter 5, *Description and Basic Inference with R*, and Chapter 6, *Group Comparisons and Statistical Modelling with R*, make those foundations executable. Chapters 7–9 then develop regression from one predictor to the regression toolbox: Chapter 7, *Simple Linear Regression*; Chapter 8, *Multiple Regression I: Conditioning, Checks, and Coordinated Questions*; and Chapter 9, *Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty*.

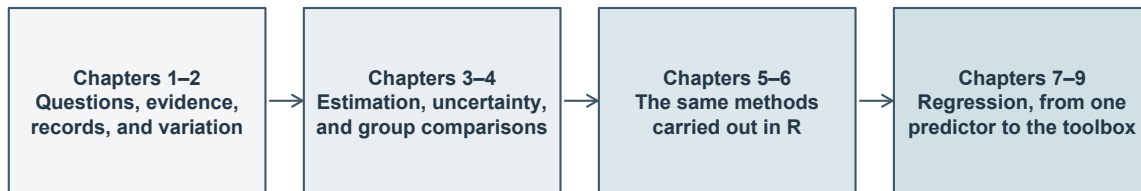


Figure 1: The route through the available chapters, from questions and records to inference, computation, and regression.

Table 1 states what each chapter is responsible for and what it builds on.

Table 1

What each available chapter is responsible for.

Chapter	What it is responsible for	What it builds on
1. The Quantitative Research Journey	Turning a concern into a question with a stated target quantity; how a study produces its evidence; how comparison, timing, sampling, and assignment support descriptive, associational, predictive, and causal answers; reading a reported result as a bounded claim	The starting point of the part

Chapter	What it is responsible for	What it builds on
2. Data, Distributions, and Variation	Reading records, identifiers, codes, and denominators; sampling schemes; distributions, their skew and tail weight, centre, spread, and groups; relationships between two variables; how a summary varies across repeated samples	Reads the dataset behind Chapter 1's account of evidence and applies its design reasoning to an interpretation
3. Estimates, Uncertainty, and Statistical Tests	The target quantity, the rule that estimates it, and the value obtained; standard errors for a mean and a difference, including the pooled standard error, and the conditions behind them; confidence intervals and confidence level; hypothesis tests, p-values, and significance; the one-sample, paired, equal-variance, and Welch t tests; the chi-square test of association and tables of three or more categorical variables; magnitude, error rates, power, and choices made after seeing the results	Turns Chapter 2's sampling distributions and standard error into inference from one observed sample
4. Group Comparisons with Analysis of Variance	One-way and two-way analysis of variance; sums of squares, degrees of freedom, mean squares, and the F test; contrasts and comparison families; interaction and simple effects, read conceptually in three- and four-way designs	Applies Chapter 2's description of group variation and Chapter 3's inference to comparisons among several groups
5. Description and Basic Inference with R	Preparing records in base R with explicit missing values; description and graphics; probability calculations; simulated sampling variation, bias, and standard error; mean intervals and z and t tests; covariance and correlation matrices and selected correlation inference; a short reproducible report	The computational route through Chapter 2 and the mean inference of Chapter 3
6. Group Comparisons and Statistical Modelling with R	The one-way and factorial analyses in R, with contrasts, comparison families, and the comparison of nested models by the F test; proportions and counts, with the effect scales that describe them; simulation, precision and power, and one scalar likelihood	Extends Chapter 4 and Chapter 3's categorical and planning lessons on Chapter 5's computational foundation

Chapter	What it is responsible for	What it builds on
7. Simple Linear Regression	A line fitted by least squares; slope, intercept, fitted values, and residuals; the matched correlation, slope, and F tests; an interval for a mean response and an interval for a new case; form, spread, influence, and supported predictor range; regression toward the mean, with Galton's heights of fathers and sons	Uses the covariance, correlation, and interval reasoning of Chapters 2, 3, and 5
8. Multiple Regression I: Conditioning, Checks, and Coordinated Questions	Several predictors and what conditioning changes; coefficient covariance and the quantities built from it; a fixed profile, standardization, and held-out prediction as different questions; several separately owed answers in one study	Extends Chapter 7's line with Chapter 5's partial correlation and Chapter 6's comparison of nested models
9. Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty	Factors, indicator coding, and declared contrasts; the analysis of covariance as a regression; nested model comparison and information criteria; polynomial terms and a changed outcome scale; heteroskedasticity-robust, cluster-robust, and bootstrap uncertainty; a randomized design whose pupils sit in schools	Collects the regression toolbox on Chapters 7–8 and supplies what the later chapters need

Each idea has one principal home, and later chapters use it rather than teach it again. Chapter 1 previews uncertainty while reading a study; Chapter 2 develops sampling distributions and the standard error through repeated samples; Chapter 3 develops their inferential use and states the conditions they require. Chapter 2 explains the descriptive ingredients of group variation; Chapter 4 develops the full analysis-of-variance decomposition and its inferential use. Where a later chapter needs an earlier idea, it recalls it briefly and then puts it to work in the new setting.

3 How to Read a Chapter and Use the Concept Reference

Chapters 1–9 share an arrangement. A **Chapter overview** box opens the chapter with the route it follows and what a reader will be able to do afterwards. Numbered sections develop the material in order. A closing **Reading Bridge** brings the principal ideas together and connects them to the next chapter, and a selective **Key terms** table gives a compact reference to the concepts explained principally there. Unnumbered material follows the numbered sections: exercises for working with the ideas, appendices holding fuller derivations and specialized extensions, and the references. Later chapters may use a different arrangement as the collection grows.

There are two sensible routes through the available chapters. The first is the chapter sequence. The second pairs statistics with computation: Chapters 2 and 3 with Chapter 5, then Chapters 3 and 4 with Chapter 6. Either way, follow a method from its data and question through its calculation to its interpretation; a result separated from the question it answers is difficult to use.

The **Part One Concept Reference** is published beside this introduction. It gathers selected concepts from Chapters 1–9 into six tables, arranged by concept group and then by term, with each term’s meaning and the place where the book explains it principally. A further table lists the procedures of Chapters 3 and 4 with what each estimates and the conditions it relies on, and a closing table records the notation those chapters use. It is a place to look something up while reading rather than a substitute for reading, because first-use explanations and worked examples stay in the chapters themselves.

Readers who would like to run R without installing anything can use the public browser practice pages that accompany Chapters 3 and 4.

4 What Comes Later

Part One continues with five further chapters. Chapter 10, *Interaction and Moderation*, develops product terms, conditional slopes, and probing within joint support. Chapter 11, *Binary Logistic Regression*, develops the Bernoulli likelihood, the logit link, and the probability, odds, and log-odds scales. Chapter 12, *Extending the GLM: Nominal, Ordinal, and Count Outcomes*, varies only the response distribution and the link for nominal, ordered, and count or rate outcomes. Chapter 13, *Prediction, Cross-Validation, and Regularization*, develops prediction as a separate aim, with held-out error and penalized fitting. Chapter 14, *Regression Pitfalls and Causal Design*, examines what can undermine a regression-based claim: reversals and Simpson’s paradox, regression to the mean, selection, measurement, dependence, and what a study’s design supplies that adjustment cannot.

Two supplementary units extend Chapter 1's argument-based reading: *How to Read a Quantitative Study as an Argument* and *Extending the Argument-Based Approach to Quantitative Studies*. Both are optional, and no numbered chapter depends on either.

Chapters 10–14 and the two supplementary units are not yet available.

Part Two of the book turns to latent-variable modeling, where the variable of primary interest is one that no column of the data set holds.